# Ancient Indian Contributions to Mathematics and Science



## Introduction

Ancient India made lasting contributions to mathematics, astronomy, medicine, metallurgy and philosophy. Indian scholars developed the decimal number system with zero, algebraic methods, trigonometry, early ideas of infinite series, theories of the atom, advanced surgery and high-quality steel, and these ideas travelled to the Arab world and then to Europe, shaping modern science. Centres of learning such as Takshashila (Taxila), Nalanda and Vikramashila attracted students from many countries. Knowledge was often recorded in Sanskrit verse, and the period from about 500 CE to 1200 CE is sometimes called the golden age of Indian mathematics and science.

## Aryabhata and the Aryabhatiya

The Aryabhatiya was written by Aryabhata, one of the earliest great Indian mathematician-astronomers. Aryabhata was born in 476 CE, probably in or near Kusumapura (Pataliputra, modern Patna), and he wrote the Aryabhatiya in 499 CE at the age of 23. The work is a short text of 121 verses organised in four sections: Gitikapada, Ganitapada (mathematics), Kalakriyapada (time reckoning) and Golapada (the sphere, astronomy). In mathematics, Aryabhata gave an accurate approximation of pi as 3.1416 (62,832 divided by 20,000) and noted that it is an approximation, gave methods for finding square and cube roots, formulas for areas, a table of sines and methods for solving indeterminate equations. In astronomy, he stated that the Earth rotates on its own axis, which causes day and night, and not that the sky revolves around the Earth; he explained solar and lunar eclipses as being caused by shadows of the Earth and the Moon, not by demons, and calculated the length of the year at about 365.358 days, very close to the true value. He also stated that the Moon and planets shine by reflected sunlight. India's first satellite, launched in 1975, was named Aryabhata in his honour. Later commentators include Bhaskara I.

## India's Contribution of Zero and the Decimal System

India's most important contribution to mathematics is zero, together with the place-value decimal system that uses ten digits, 0 to 9. The concept of shunya (void or nothing) existed in Indian philosophy, and Indian mathematicians turned it into a number that could be used in calculation, as well as a placeholder symbol in the place-value system. The Bakhshali manuscript, found in 1881 in a village near Peshawar, contains hundreds of zeros written as dots; radiocarbon dating done at Oxford University and announced in 2017 placed parts of it in the third or fourth century CE, making it the oldest known record of a zero used as a symbol in South Asia. In 628 CE, Brahmagupta (598 to 668 CE), in his Brahmasphutasiddhanta, gave the first clear rules for arithmetic with zero and negative numbers, such as that a number added to zero is unchanged and that a debt subtracted from zero is a fortune. The Indian numerals were taken up by Arab mathematicians such as Al-Khwarizmi, who wrote about them in the 9th century, and then taken to Europe in the Middle Ages, notably by Fibonacci's Liber Abaci in 1202, so they are called Hindu-Arabic numerals. Without zero, modern computing, which is built on binary digits, would be impossible. Pingala, in the Chandahshastra around the 3rd or 2nd century BCE, described a binary system for poetic metres.

## Other Mathematicians and Astronomers

Baudhayana's Sulba Sutras, from about 800 BCE, state the so-called Pythagorean theorem long before Pythagoras and give an approximation for the square root of 2. Brahmagupta also gave a formula for the area of a cyclic quadrilateral and solved quadratic equations. Varahamihira (505 to 587 CE) wrote the Brihat Samhita and the Pancha Siddhantika and made astronomical and trigonometric tables. Bhaskara II (1114 to 1185 CE) wrote the Lilavati, a famous book of arithmetic problems, and the Siddhanta Shiromani, and discussed ideas close to differential calculus. Madhava of Sangamagrama (about 1340 to 1425 CE) in Kerala founded the Kerala School of astronomy and mathematics and discovered infinite series for sine, cosine, arctangent and pi, about two centuries before Newton and Leibniz.

## Medicine: Who Was Sushruta?

Sushruta, who lived around the 6th century BCE (some scholars place him a little later), in Kashi (Varanasi), is called the "Father of Surgery" and also the father of plastic surgery. He is the author of the Sushruta Samhita, one of the foundational texts of Ayurveda, which describes more than 300 surgical procedures, about 120 surgical instruments, such as scalpels, forceps and probes, and more than 1,100 diseases. He explained how to perform rhinoplasty, rebuilding a nose with a skin flap from the cheek, cataract surgery by couching, the removal of bladder stones, fracture setting, and suturing wounds, and stressed cleanliness, sterilising instruments and using wine or herbs to reduce pain. He recommended learning anatomy by dissecting a body that had been soaked in water. Charaka, the author of the Charaka Samhita, is regarded as the father of Indian medicine, and focused on internal medicine, diagnosis, diet and ethics. Ayurveda, Siddha, Unani and Yoga are recognised traditional systems under the Ministry of AYUSH, and Siddha is especially strong in Tamil Nadu, with Siddhars such as Agastya and Bogar.

## Physics, Chemistry and Metallurgy

Kanada, in the Vaisheshika school around the 6th century BCE, taught that all matter is made of tiny indivisible particles called anu (atoms) and wrote about motion and light. Indian metallurgists made wootz steel, a high-carbon steel exported to the Middle East where it was made into the famous Damascus swords. The Iron Pillar of Delhi, about 1,600 years old and 7 metres tall, has resisted rust. Zinc was distilled on an industrial scale at Zawar in Rajasthan. Ancient Indian textile dyeing, cotton cultivation, shipbuilding and the Harappan town planning also show scientific skill. The Jantar Mantar observatories built by Maharaja Jai Singh II in the 18th century measure time and astronomical positions.

## Modern Echoes: Ramanujan and Tamil Nadu

The tradition continued into modern times. Srinivasa Ramanujan, born on 22 December 1887 in Erode, Tamil Nadu, and raised in Kumbakonam, discovered thousands of formulae in number theory, infinite series and continued fractions with almost no formal training, and was elected a Fellow of the Royal Society in 1918. The number 1729, which he called interesting because it is the smallest number expressible as the sum of two cubes in two ways, is known as the Hardy-Ramanujan number. India celebrates 22 December as National Mathematics Day. C. V. Raman of Tamil Nadu won the Nobel Prize in Physics in 1930, S. Chandrasekhar the Nobel Prize in 1983 and Venkatraman Ramakrishnan, who was born in Chidambaram, Tamil Nadu, the Nobel Prize in Chemistry in 2009.

## Challenges and Future

Historians debate dates and origins, and researchers work to preserve manuscripts. Today India promotes the study of its heritage through the Indian Knowledge Systems initiative along with modern science.

## Quick Facts

- The Aryabhatiya was written by Aryabhata in 499 CE.
- India gave the world zero and the decimal place-value system; Brahmagupta gave rules for zero in 628 CE.
- Sushruta, author of the Sushruta Samhita, is called the Father of Surgery.
- Madhava of Kerala discovered infinite series for sine, cosine and pi centuries before Newton.
- Srinivasa Ramanujan was born in Erode, Tamil Nadu, in 1887.